

Syntax

- The **signature** $\mathcal{L}$ now consists of:
 - set of sorts, denoted by α, β
 - constants, denoted by a, b, c , each having a sort $c : \alpha$
 - function symbols, denoted by f, g , each having a type $\alpha_1 \times \dots \times \alpha_n \mapsto \alpha$
 - predicate symbols, denoted by p, q , each having a type $\alpha_1 \times \dots \times \alpha_n$

Quantifiers:

- Given a formula G , a variable x , then $\mathcal{Q}x : \alpha A$ is a quantified formula with $\mathcal{Q} \in \{\forall, \exists\}$
- A formula is called quantifier-free (or ground) if it contains no occurrences of quantifiers

Semantics

- The **interpretation** I maps:
 - Each sort α to a non-empty set $I(\alpha)$, called domain of the sort
 - Each constant $c : \alpha$ to $c' \in I(\alpha)$
 - Each variable $x : \alpha$ to $x' \in I(\alpha)$
 - Each function symbol $f : \alpha_1 \times \dots \times \alpha_n \mapsto \alpha$ to a function $f' : I(\alpha_1) \times \dots \times I(\alpha_n) \mapsto I(\alpha)$
 - Each predicate symbol $p : \alpha_1 \times \dots \times \alpha_n$ to a relation $p' \subseteq I(\alpha_1) \times \dots \times I(\alpha_n)$

- Given I and a term t of a sort α , define the element $t' \in I(\alpha)$ as follows:

- For constants and variables $t' = I(t)$
- $f(t_1, \dots, t_n)' = f'(t_1', \dots, t_n')$

- Given I and a formula G , define the boolean value as follows:

- For an atomic formula $p(t_1, \dots, t_n)$, define $p(t_1, \dots, t_n)' = 1 = (t_1', \dots, t_n') \in p'$
- $G = (Q_1 \circ Q_2)' = 1$ if $Q_1' \circ Q_2'$ is satisfied

Quantifiers

- Given a set sequence of variables $\bar{x}$, two interpretations of the same signature are $\bar{x}$ -variant if they coincide on all symbols and variables not $\bar{x}$.
- $(\forall xA)' = 1$ if for all $\bar{x}$ -variants I' of I we have $(A)' = 1$
- $(\exists xA)' = 1$ if for some $\bar{x}$ -variant I' of I we have $(A)' = 1$

Axiomatic Interpretations

Restrict interpretations to those which satisfy the following set of formulas (so called axioms):

- reflexivity : $x = x$
- symmetry : $x = y \Rightarrow y = x$
- transitivity : $x = y \wedge y = z \Rightarrow x = z$
- congruence axioms for each function symbol f :

$$\left(\bigwedge_{i=1}^n x_i = y_i \right) \rightarrow f(x_1, \dots, x_n) = f(y_1, \dots, y_n)$$

- congruence axioms for each predicate symbol p :

$$\left(\bigwedge_{i=1}^n x_i = y_i \wedge p(x_1, \dots, x_n) \right) \rightarrow p(y_1, \dots, y_n)$$

Interpretation Classes

Restrict interpolations to those which satisfy certain properties

Example :

- Given the formula $f(x) = f(x+1) \wedge f(a) \neq f(b)$
- 1. We consider the class of interpretations I where I interprets $+$ and 1 in the standard way over the integers. The formula is satisfiable in:

$$I(a) = 1, I(b) = 2, I(x) = 3$$
$$f(x) = \begin{cases} 3 & \text{if } x = 4 \\ x & \text{else} \end{cases}$$

- 2. We consider the class of interpretations of example 1. but where $f(x) = f(x+1)$ is valid in I .

The formula is not satisfiable as $f(x) = f(x+1)$ would impose that f maps all variables to the same value, in which case $f(a) \neq f(b)$ can not hold.

Satisfiability and Interpretation - Examples

- Given the formula $x = y + 1$ over integers, where x, y are variables, $+$, 1 are interpreted in the standard way over integers and the interpretation $I(x) = I(y) = 0$. It then holds that:
 - $I \models x = y + 1$
 - $x = y + 1$ is satisfiable in I under the $\bar{x}$ -variant: $I'(x) = 1, I(y) = 0$
 - $x = y + 1$ is not valid in I , since $I \not\models x = y + 1$

- Given the formula $z \neq a \rightarrow z = b$, where $a : \alpha, b : \alpha$ are constants and $z : \alpha$ is a variable. This formula is valid under the class of interpretations I for which it holds that:

$$-D_a^I = \{0, 1\}, I(a) = 0, I(b) = 1$$

- Given the formula $y = a \rightarrow y = b$, where a, b are constants and y is a variable. This formula is valid under the class of interpretations I for which it holds that :
 - $I(a) = I(b)$

Theory

- A theory is defined as
 - a signature Σ_T
 - a set of formulas $\mathcal{A}_T$ over Σ_T , called axioms of T

Unless specified otherwise, theories are quantifier-free

- An interpretation in which $\mathcal{A}_T$ are valid is called a T -interpretation
- If a formula G is valid in every T -interpretation, then G is T -valid (denoted by $T \models G$)

Theory of equality T_E :

- $\Sigma_E = \{a, b, \dots, f, g, \dots, p, \dots\}$
- $A_E = \left\{ \begin{aligned} &x = x, x = y \rightarrow y = x, x = y \wedge y = z \rightarrow x = z, \\ &\bigwedge_{i=1}^n y_i = x_i \rightarrow f(x_1, \dots, x_n) = f(y_1, \dots, y_n), \\ &\bigwedge_{i=1}^n y_i = x_i \wedge p(x_1, \dots, x_n) \rightarrow p(y_1, \dots, y_n) \end{aligned} \right\}$

A binary relation R over a set S is called a congruence relation, if:

- R is reflexive, transitive, symmetric and
- $\bigwedge_{i=1}^n y_i R x_i \rightarrow f(x_1, \dots, x_n) R f(y_1, \dots, y_n)$

The congruence class of $t \in S$ under R is $[t]_R = \{t' \in S \mid t R t'\}$,

for which holds:

- $\bigcup_{[t]_R} [t]_R = S$
- $[t_1]_R \neq [t_2]_R \rightarrow [t_1]_R \cap [t_2]_R = \emptyset$

The congruence closure R^c of R is the relation for which holds:

- $R \subseteq R^c$
- $\forall R' : R \subseteq R' \rightarrow (R^c = R' \vee R^c \subseteq R')$

Congruence Closure Algorithm for T_E

```
procedure CongruenceClosure(F)
input: F is  $s_1 = t_1 \wedge \dots \wedge s_m = t_m \wedge s_{m+1} \neq t_{m+1} \wedge \dots \wedge s_m \neq t_m$ 
output: satisfiable or unsatisfiable
parameters: function subterm.set
begin
   $S_T := \text{subterm.set}(F)$ 
   $R := \{sRt \mid s \in S_T, t \in S_T\}$  and  $[s]_R := \{s\}$ , defining the partition  $\{[s]_R \mid s \in S_T\}$ 
  for every  $s_i = t_i$  in  $F$ , merge  $[s_i]_R$  and  $[t_i]_R$  by
  forming the union  $[s_i]_R \cup [t_i]_R$ 
  propagate the new congruences that arise in this union (function congruence)
  if  $s_i R t_i$  for any  $i \in \{n+1, \dots, m\}$  then return unsatisfiable
  else return satisfiable end
```

Example

Given the formula $F : f(a, b) = a \wedge f(f(a, b), b) \neq a$

We proceed as follows:

- 1. Subterm Set $S_F = \{a, b, f(a, b), f(f(a, b), b)\}$
- 2. Initial partition $P = \{\{a\}, \{b\}, \{f(a, b)\}, \{f(f(a, b), b)\}\}$

- 3. Given $f(a, b)Ra$, we merge the partition :

$$P = \{\{a, f(a, b)\}, \{b\}, \{f(f(a, b), b)\}\}$$

- 4. Given $f(a, b)Ra \wedge bRb \rightarrow f(f(a, b), b)Rf(a, b)$:

$$P = \{\{a, f(a, b), f(f(a, b), b)\}, \{b\}\}$$

- 5. Due to the clause $f(f(a, b), b) \neq a$, formula F is T_E -unsatisfiable, since $f(f(a, b), b) \in [a]_R \iff f(f(a, b), b) R a \in R$, but the formula states, that $f(f(a, b), b) \neq a$

DPPL(T) - For Non-Unit Clauses in T_E

Given a formula $G := (a = b \vee a = c) \wedge \dots$ then G contains a non-unit clause and can thus not be proven with congruence closure. Instead, use DPPL $_T$:

```
procedure DPPL $_T$ (F)
input: a CNF formula F in the decidable theory T
output: satisfiable or unsatisfiable
parameters: functions propositional.abstraction, decision.procedure $_T$ 
begin
   $\bar{S} := \text{propositional.abstraction}(F)$ 
  while true do
    if DPPL(S) = unsatisfiable then return unsatisfiable
    if DPPL(S) = satisfiable then
       $L := \{L_1, \dots, L_n\}$ , s.t. for the model I of I we have  $I \models S$ 
      if decision.procedure $_T$ (L) = satisfiable then return satisfiable
      if decision.procedure $_T$ (L) = unsatisfiable then
         $S := S \cup \{\neg L_1 \vee \dots \vee \neg L_n\}$ 
      end do
    end do
  end
```

Example

Given the formula F :
 $(a = b \vee a = c) \wedge f(a) = a \wedge f(f(b)) \neq a \wedge f(c) \neq a$

- 1. Add propositional symbol for every theory atom:

$$p_1 : a = b, p_2 : a = c, p_3 : f(a) = a,$$
$$p_4 : f(f(b)) \neq a, p_5 : f(c) = a$$
$$F_P = (p_1 \vee p_2) \wedge p_3 \wedge \neg p_4 \wedge \neg p_5$$

- 2. If F_P is unsat, then so is F

- 3. If F_P is sat, then we have literals $L_1, \dots, L_n$

of I for which holds: $I \models F_P$

$$\text{e.g. } p_1, \bar{p}_2, \bar{p}_3, p_4, \bar{p}_5$$

- 4. Use clauses corresponding to literals and check with congruence closure

$$F' = a = b \wedge a \neq c \wedge f(a)$$
$$\xrightarrow{\text{simp } \bar{p}_2}$$

- 5. If F' is T -sat, terminate, else define $F_P = F_P \wedge (\neg L_1 \vee \dots \vee \neg L_2)$ and go to 2.

Theory of arrays T_A :

- $\Sigma_A = \{\text{read}, \text{write}\}$
- where read is a binary function and write is ternary function
- $A_A = A_E \cup \left\{ \begin{aligned} &x = y \rightarrow \text{read}(\text{write}(A, x, v), y) = v, & \text{r-o-w}_1 \\ &x \neq y \rightarrow \text{read}(\text{write}(A, x, v), y) = \text{read}(A, y) & \text{r-o-w}_2 \end{aligned} \right\}$

- $\text{read}(A, x)$ denotes the value of array A at pos. x
- $\text{write}(A, x, v)$ denotes the modified array A with value v at pos. x

Given a function F with write -terms $\text{read}(\text{write}(A, x, v), y)$, $F[F']$ denotes the formula where the write -term has been replaced with F'

Congruence Closure Algorithm for T_A

T_A -satisfiability of F is reduced to T_E -satisfiability:

- If F contains write -terms $\text{read}(\text{write}(A, x, v), y)$, use r-o-w axioms:

- 1. $F_1 : x = y \wedge F[v]$
- 2. $F_2 : x \neq y \wedge F[\text{read}(A, y)]$
- If both are unsat, then F is unsat

- If F doesn't contain write -terms:

- Use fresh function symbol f_A for array A
- Replace all $\text{read}(A, x)$ with $f_A(x)$

Return T_E satisfiability of modified formula

Combination of Theories

Given:

- Given a set of theories $T_1, \dots, T_n$ in disjoint $\Sigma_1, \dots, \Sigma_n$ and its axioms $\bigcup_{i=1}^n \mathcal{A}_{T_i}$

- Quantifier-free conjunction F of literals in $\bigcup_{i=1}^n \Sigma_i$

Checking if F is satisfiable in a T :

- Transform $F : \bigwedge_{i=1}^n F_i$ with F_i being a T_i formula, where $F_1, \dots, F_n$ are linked by variables

Example:

- Given T_A, T_E and $F : (y = x + 2) \wedge (f(\text{read}(\text{write}(a, x, 3), y - 2)) \neq f(y - x + 1))$, we split:
 $F : (y = x + 2) \wedge f(c_4) \neq f(c_1)$
- Shared equalities are derived e.g
- $-y = x + 2 \wedge c_2 = y - 2$
 $\implies c_2 = x$

$-c_4 = \text{read}(\text{write}(a, x, c_3), c_2) \wedge c_2 = c_x$ $\xrightarrow{\text{r-o-w}_2 \text{ axiom}}$ $\implies c_4 = c_3$	<table><tr><th>Linear Arithmetic</th><th>Uninterpreted Functions</th></tr><tr><td>$y = x + 2$ $c_1 = y - x + 1$ $c_2 = y - 2$ $c_3 = 3$</td><td>$f(c_4) \neq f(c_1)$</td></tr><tr><th>Arrays</th><td></td></tr><tr><td>$c_4 = \text{read}(\text{write}(a, x, c_3), c_2)$</td><td>$c_2 = x$ $c_4 = c_3$ $c_1 = c_2$</td></tr><tr><th>Shared Equalities</th><td></td></tr><tr><td></td><td>$f(c_4) \neq f(c_1) \implies c_4 \neq c_1$ $c_4 = c_3$ $c_3 = c_1$ $\implies \text{Contradiction! The formula is unsat}$</td></tr></table>	Linear Arithmetic	Uninterpreted Functions	$y = x + 2$ $c_1 = y - x + 1$ $c_2 = y - 2$ $c_3 = 3$	$f(c_4) \neq f(c_1)$	Arrays		$c_4 = \text{read}(\text{write}(a, x, c_3), c_2)$	$c_2 = x$ $c_4 = c_3$ $c_1 = c_2$	Shared Equalities			$f(c_4) \neq f(c_1) \implies c_4 \neq c_1$ $c_4 = c_3$ $c_3 = c_1$ $\implies \text{Contradiction! The formula is unsat}$
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Split Reasoning Algorithm

```
procedure SeparatingReasoning(F)
input: formula  $F \in T_1 \cup \dots \cup T_n$ 
output: formulas  $F_1 \in T_1, \dots, F_n \in T_n$  s.t.
 $F$  and  $F_1 \wedge \dots \wedge F_n$  are equisatisfiable
assumptions: theory signatures  $\Sigma_1, \dots, \Sigma_n$  are disjoint
parameters: function head(T) returning the root symbol of a term t
begin
  repeat as long as possible
    if  $F \in \Sigma_i$  and head(F)  $\in \Sigma_i$  with  $i \neq j$ :
      rewrite  $F[f(t_1, \dots, t_n)]$  into  $F[f(c_1, \dots, c_n)] \wedge c = t$ ,
      where  $c$  is a fresh new variable
    if  $p \in \Sigma_i$  and head(p)  $\in \Sigma_i$  with  $i \neq j$ :
      rewrite  $F[p(t_1, \dots, t_n)]$  into  $F[p(c_1, \dots, c_n)] \wedge c = t$ ,
      where  $c$  is a fresh new variable
    if head(s)  $\in \Sigma_i$  and head(t)  $\in \Sigma_j$  with  $i \neq j$ :
      rewrite  $F[s = t]$  into  $F[s = c] \wedge c = t$ ,
      where  $c$  is a fresh new variable
    end repeat
  return modified  $F$  as  $F_1 \wedge \dots \wedge F_n$ , with each  $F_i \in T_i$ 
```

Not all theories can be easily combined this way

Convex Theory

A formula $F \in \mathcal{T}$ as a conjunction of

T -literals is convex if:

$$-T \models \left(F \rightarrow \bigvee_{j=1}^k (u_j = v_j) \right) \implies T \models F \rightarrow u_j = v_j$$

for some $j \in \{1, \dots, k\}$

Example

T_A e.g. is not convex, T_E e.g. is convex

For T_A , let $F : \text{write}(A, x, v_1) = \text{write}(A, y, v_2)$

By $\mathcal{A}_A$ it follows that $x = y \vee v_1 = v_2$, but the same does not hold for the separate T_A literals.

Stably infinite theories

A theory T_i is called stably infinite:

- For every T satisfiable formula $F \in T$, there exists a T -interpretation I :
 - $I \models F$
 - I has a domain of infinite cardinality

T_E, T_A, T_Z are stably infinite

Nelson Oppen method

Given $T_1, \dots, T_n$ decidable, quantifier-free theories with disjoint signature and which are stably infinite the Nelson Oppen Method for **convex** theories finds new equalities $u = v$ for shared u, v . Pass equality to other theories.

```
procedure NelsonOppenConvex(F)
input: conjunctive formula  $F \in T_1 \cup \dots \cup T_n$ 
output: satisfiable or unsatisfiable
assumptions: Nelson-Oppen restrictions over  $T_i$  hold
assumptions:  $F_i$  are convex
begin
  1.  $F_1 \wedge \dots \wedge F_n := \text{SeparatingReasoning}(F)$ 
  2. Run  $T_i$ -solver on  $F_i$ . If a solver returns unsatisfiable, return unsatisfiable
  3. If there are shared variables  $u, v$  s.t.  $F_i \rightarrow u = v$  then
      $F_i := F_i \wedge u = v$ 
     go to step 2
  return satisfiable
end

procedure NelsonOppenNonConvex(F)
input: conjunctive formula  $F \in T_1 \cup \dots \cup T_n$ 
output: satisfiable or unsatisfiable
assumptions: Nelson-Oppen restrictions over  $T_i$  hold
assumptions:  $F_i$  are non-convex
begin
  1.  $F_1 \wedge \dots \wedge F_n := \text{SeparatingReasoning}(F)$ 
  2. Run  $T_i$ -solver on  $F_i$ . If a solver returns unsatisfiable, return unsatisfiable
  3. If there are shared variables  $u, v$  s.t.  $F_i \rightarrow u = v$  then
      $F_i := F_i \wedge u = v$ 
     go to step 2
  4. If  $F_i \rightarrow \bigvee_k u_k = v_k$  for shared  $u_k, v_k$ , then
     if NelsonOppenNonConvex( $F_1 \wedge \dots \wedge F_n \wedge u_k = v_k$ ) is satisfiable, for any  $k$ 
     then return satisfiable
     else return unsatisfiable
  return satisfiable
end
```

SMT